

$\Rightarrow$ Remainder (%)

$$10 \% 4 \Rightarrow 2$$

$$\text{quiz 1} \Rightarrow 99 \% 8 \Rightarrow 3$$

$a \% b \Rightarrow$ remainder when a is divided by b .

$$\text{Remainder} = \text{Dividend} - \underbrace{\text{Quotient} \times \text{Divisor}}_{\substack{\downarrow \\ \text{max multiple of} \\ \text{divisor} \leq \text{dividend}}}$$

$$99 \% 8$$

$$\begin{aligned} r &= 99 - 96 \\ &= 3. \end{aligned}$$

$$10 \% 4$$

$$\begin{aligned} r &= 10 - 8 \\ &= 2, \end{aligned}$$

$$\text{quiz 2} \Rightarrow -40 \% 7 \Rightarrow -40 - (\text{max multiple of } 7 \leq -40)$$

$$\Rightarrow -40 - (-42)$$

$$\Rightarrow -40 + 42$$

$$\Rightarrow 2$$

$$\text{Ques 3} \Rightarrow -60 \% 9 \Rightarrow -60 - (\text{max multiple of } 9 \leq -60)$$

$$\Rightarrow -60 - (-63)$$

$$\Rightarrow -60 + 63$$

$$\Rightarrow \underline{\underline{3}}$$

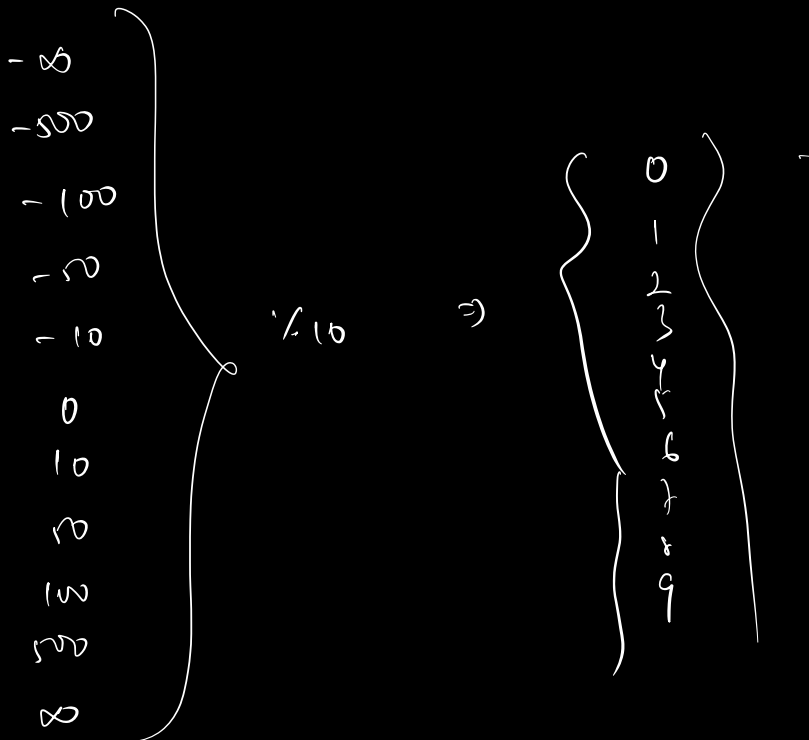
C/C++/Java

Python

$$-40 \% 7 \Rightarrow -5 + 7 \longrightarrow 2$$

$$-60 \% 9 \Rightarrow -6 + 9 \longrightarrow 3$$

Why ?



Quiz 4 $\Rightarrow$ max^m value of $a \% m \Rightarrow [0, m-1]$

$$2 \% 4 = 2$$
$$10 \% 100 = 10$$

$$\downarrow$$
$$\text{max}^m \Rightarrow \underline{m-1}$$

$\Rightarrow$ Modular Arithmetic :

1) $(a+b) \% m \Rightarrow [a \% m + b \% m] \% m$

2) $a \% m \Rightarrow (a+m) \% m$

$$\downarrow$$
$$(a \% m + m \% m) \% m$$
$$\Rightarrow (a \% m + 0) \% m$$
$$\Rightarrow (a \% m) \% m$$
$$\downarrow$$
$$[0, m-1]$$
$$\Rightarrow \underline{a \% m}$$

3) $(a \times b) \% m \Rightarrow (a \% m \times b \% m) \% m$

4) $(a-b) \% m \Rightarrow (a \% m - b \% m + m) \% m$

ex $\Rightarrow a=8, b=4, m=5$

$$(8-4) \% 5$$

$$\Rightarrow 4 \% 5$$

$$\Rightarrow 4$$

$$\downarrow$$
$$(8 \% 5 - 4 \% 5 + 5) \% 5$$
$$\Rightarrow (3 - 4 + 5) \% 5$$

$$\Rightarrow (1 + 5) \% 5$$

$$\Rightarrow 4 \% 5 \Rightarrow 4$$

ex $\Rightarrow$ $A = 13$ $M = 2, 3, 6$ $Op = \underline{\underline{3}}$
 $B = 7$

brute force

for ($M=2$ to $M=9$) {

check ($A \% M, B \% M$)

$\Rightarrow \frac{O(A)}{TC}$

}

$$A \% M = B \% M$$

$$\Rightarrow A \% M - B \% M = 0$$

+M on both sides

$$\Rightarrow A \% M - B \% M + M = M$$

+M on both sides

$$\Rightarrow (A \% M - B \% M + M) \% M = M \% M$$

$$\Rightarrow (A \% M - B \% M + M) \% M = 0 \longrightarrow \underline{\underline{(4)}}$$

$$\Rightarrow (A - B) \% M = 0$$

$\hookrightarrow$ possible values of

M are all factors of (A-B)

except 1.

$\downarrow$
sqrt()

$$(A-B) \% M = 0 \Rightarrow (A \% M - B \% M + M) \% M$$

TC $\Rightarrow O(\sqrt{A-B})$
 SC $\Rightarrow O(1)$

Q2, Given N array elements, calculate no. of pairs (i, j) such that $\{[arr[i] + arr[j]] \% M = 0\}$ M is given.

Note: $(i \neq j)$, & $\{pair(i, j) \text{ is same as } pair(j, i)\}$

ex $\Rightarrow$ $arr[6] \Rightarrow$

0	1	2	3	4	5
4	7	6	5	5	3

 $M = 3$

pair	i	j	arr[i]	arr[j]	sum % M
	0	3	4	5	9 % 3
	0	4	4	5	9 % 3
	1	3	7	5	12 % 3
	1	4	7	5	12 % 3
	2	5	6	3	9 % 3

} $\underline{5}$
 $\text{O/p} = \underline{5}$

ex 2 $\Rightarrow$ $arr \Rightarrow [13, 14, 22, 3, 22, 19, 16]$

Brute force

$\Rightarrow$ try all pairs

for ($i=0; i < N; i++$) {

for ($j=i+1; j < N; j++$) {

if ($(arr[i] + arr[j]) \% M == 0$)

count++

}

TC $\Rightarrow O(N^2)$
SC $\Rightarrow O(1)$

Ans:

$$(arr[i] + arr[j]) \% M = 0$$

$$\Rightarrow [arr[i] \% M + arr[j] \% M] \% M = 0$$

M=10

arr = 17 2 5 4 6 23 13 26 14 18 15 30 35

M=10

arr = 7 2 5 4 6 3 3 6 4 8 5 0 5

[arr[i] + arr[j]] % M

arr[12] = {24, 16, 8, 7, 17, 23, 35, 20, 26, 28, 34, 21}

M=6

remainder

→

0 1 2 3 4 5

no. of elements

→

1	1	3	1	3	3
---	---	---	---	---	---

having remainder i

24 + 1-5 + 0 + 3

total pairs = 9 3 0 0

= 12

arr = { _____ } M=6

remainder

⇒

ind is remainder

→

0 1 2 3 4 5

M=6

array

↙

4 2 3 5 2 3

count of elements having remainder i

$$\begin{aligned}
 \text{total pairs} &= {}^0C_2 + {}^{1-5}C_2 + {}^{2-4}C_2 + {}^3C_2 \\
 &= \frac{4(4-1)}{2} + 6 + 6 + 5(5-1) \\
 &= 6 + 6 + 6 + 10 \\
 &= 28 \text{ pairs}
 \end{aligned}$$

$N C_2$
 $\downarrow$
 no. of
 pairs of
 size 2
 out of
 no elements
 $N C_2 = \frac{N(N-1)}{2}$

Step 1 $\rightarrow$ remainder array

$rm[M]$

for (int i=0; i<N; i++) {

$rm[arr[i] \% M]++$

}

$$ans = {}^{{rm[0]}}C_2 \rightarrow \frac{{rm[0]({rm[0]-1})}}{2}$$

i=1; j=M-1

Step 2 while (i<j) {

$ans = ans + \{rm[i] \times rm[j]\}$

i++; j--;

}

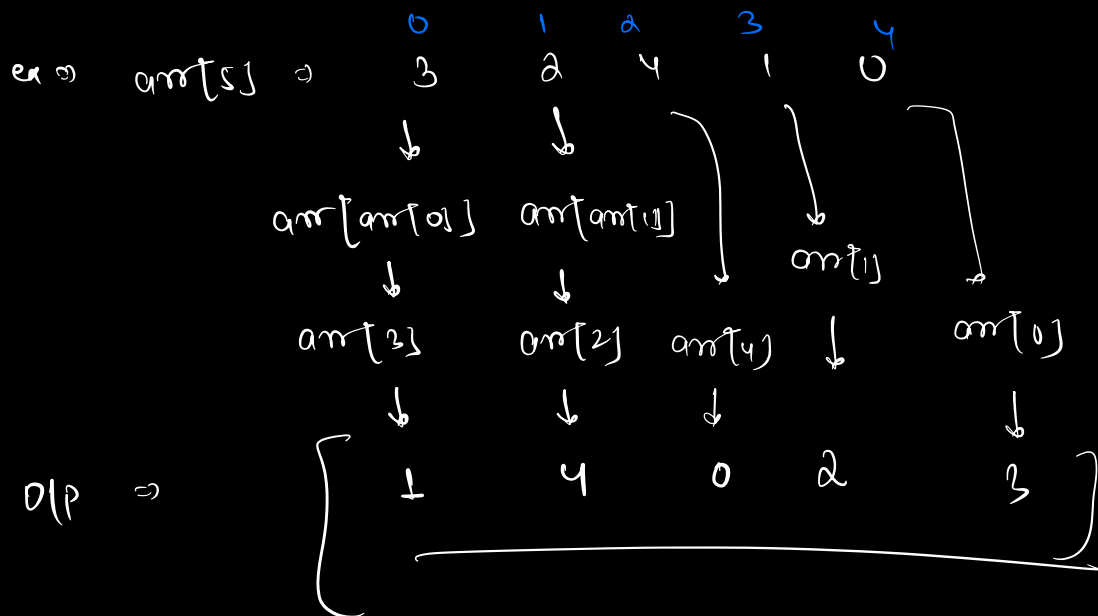
Step 3 : if ($M \% 2 \neq 0$) {

ans = ans + $\frac{arr[M/2] \times (arr[M/2] - 1)}{2}$;
}

$TC \Rightarrow O(N)$
 $SC \Rightarrow O(1)$

Q 3. Given an array of all distinct integers in the range of $\{0 \leq arr[i] \leq N-1\}$ N is the size of array.

Replace $arr[i]$ with $arr[arr[i]]$



arr[] $\Rightarrow$

	0	1	2	3	4	5	6
	3	1	4	6	5	0	2
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
$\Rightarrow$	6	1	5	2	0	3	4

Ques $\Rightarrow$

arr[] $\Rightarrow$ { 1, 6, 3, 5, 4, 2, 0 }

	0	1	2	3	4	5	6
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
$\Rightarrow$	6	0	5	2	4	3	1

$\Rightarrow$ No extra space

Day	Hours		D	T
0	0	23 hrs $\rightarrow$	0	23 hrs.
		46 hrs $\rightarrow$	1	22 hrs
		100 hrs $\rightarrow$	4	4 hrs.
		125 hrs $\rightarrow$	5	5 hrs.

x hrs $\rightarrow$ $x/24$ day $\quad x \% 24$ hrs

$\swarrow$ $\searrow$

Quotient $\quad$ remainder

$x/24 \quad \quad x \% 24$

arr[i]

$\rightarrow$ old value $\rightarrow /N \rightarrow$ old value

$\rightarrow$ new value $\rightarrow \% N \rightarrow$ new value

arr[5] $\Rightarrow$

0	1	2	3	4
3	2	4	1	0

$N=5$

multiply x5 $\rightarrow$

15	10	20	5	0
(3x5)	(2x5)	(4x5)	(1x5)	(0x5)

org. value $\rightarrow$

$15/3 = 3$	$10/5 = 2$	$20/5 = 4$	$5/5 = 1$	$0/5 = 0$
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arr[org] = val arr[3] = 1 arr[2] = 4 arr[4] = 0 arr[1] = 2 arr[0] = 3

add val to current $\rightarrow$

16	14	20	7	3
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final array

org value $\rightarrow$

3	2	4	1	0
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new val $\% 5 \Rightarrow$

1	4	0	2	3
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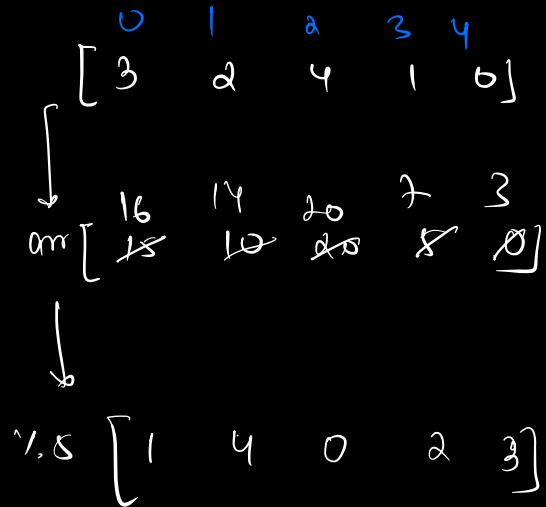
$\Rightarrow$ updating the array

```
for (i=0; i<N; i++) {
    arr[i] = arr[i] x N
}
```

	0	1	2	3	4
	1	4	0	2	3
	3	2	4	1	0
arr[15]	10	20	5	0	

⇒ getting both values

```
for (i = 0; i < N; i++) {
    org = arr[i] / N
    newV = arr[org] / N
    arr[i] = arr[i] + newV
}
```



```
⇒ O/p : for (i = 0; i < N; i++) {
    arr[i] = arr[i] / N
}
```

when arr

$$\begin{aligned} TC &= O(N) \\ SC &= O(1) \end{aligned}$$

$$\underbrace{[a \times m]}_{\text{}} \times M = 0$$

$$\underbrace{[arr[i] \times N]}_{\text{}} \times N = 0$$

$$\left[\underbrace{[0 \times m]}_{\text{}} + x \right] \times M = x$$

$$\underline{\underline{x < M}}$$

$$\left[arr[i] \times N + \underline{\underline{newV}} \right] \times N$$

$$\underline{\underline{newV < N}}$$

$$\geq \underline{\underline{newV \neq 0}}$$